

Lifting Line Group Assignment

ROTOR & WAKE AERODYNAMICS

AE4135

Authors:



Szczepan Zygmunt Letkiewicz (5486971)

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1 Introduction

One of the classical methods used to evaluate the aerodynamic performance of a wing is Lifting Line Theory. This technique models a three-dimensional wing—or in this case, a wind turbine blade—as a line of bound vorticity at the quarter chord line. This approach stems from potential flow theory and offers a computationally efficient way to estimate the spanwise lift distribution and induced velocities for high-aspect-ratio blades.

In this report, a frozen vortex wake model based on lifting line theory is implemented and validated. The methodology involves discretizing the rotor blade span into control points located at the quarter chord and forming horseshoe vortices to simulate the bound and trailing vorticity. The induced velocities at these control points are calculated using the Biot-Savart law, and the resulting circulation is solved iteratively until convergence is achieved. The model is applied to both single and dual-rotor configurations to assess their interaction with each other.

A secondary method based on the impermeability boundary condition is also explored as an alternative to the iterative approach. Furthermore, the accuracy of the lifting line model is evaluated through comparison with BEM results. Finally, a sensitivity study is also performed with particular attention paid to parameters such as wake length, blade discretization, and azimuthal resolution. This method enables a comprehensive assessment of the lifting line method's capabilities and limitations in simulating rotor aerodynamics.

2 Flowchart

The frozen wake lifting line theory is an aerodynamic model used to calculate the circulation distribution along a blade. In this report, the approach to creating the control points and filaments used to assemble horseshoe vortices is completed on one blade and then rotated by 120° in the azimuthal direction. Control points are defined in the middle of each radial element at the quarter chord. The wake is modeled using five filaments in total. The basic horseshoe vortex consists of one bound filament along the blade span (F1) and two trailing filaments (F2 and F3). To represent the frozen wake, two additional filaments (F4 and F5) are attached to the trailing filaments. These capture the displacement of the wake in the x, y, z directions, as illustrated in [Figure 1](#).

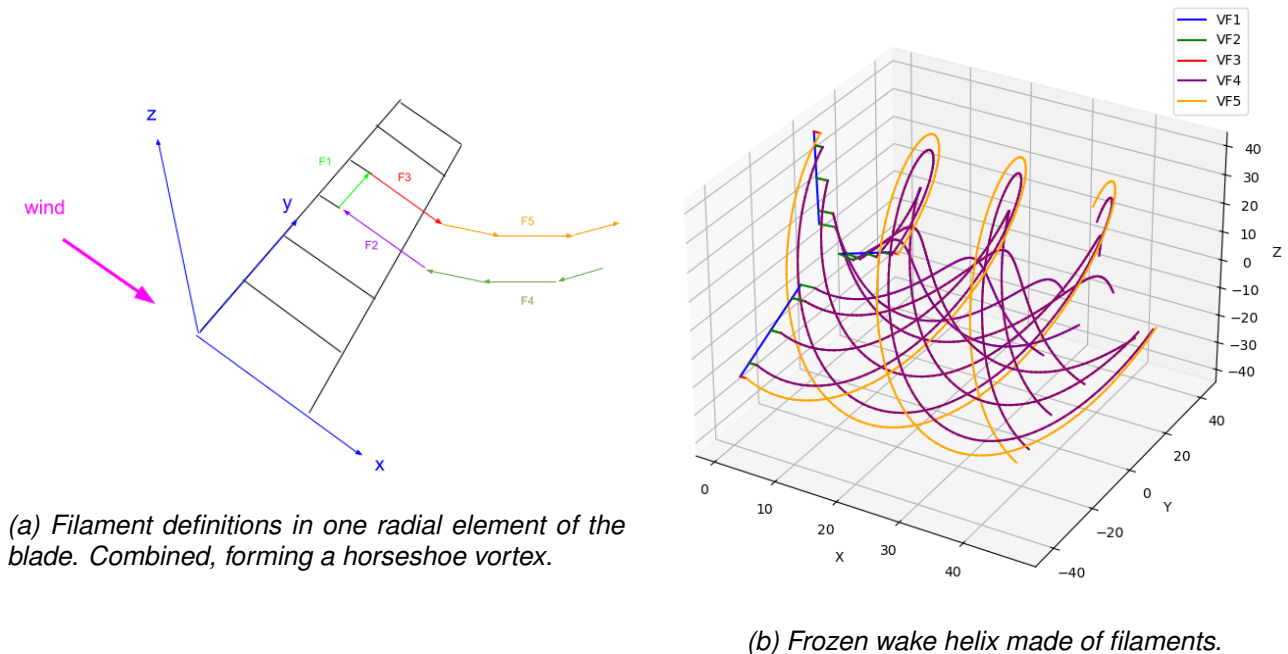


Figure 1: Lifting Line Filaments

These vortices are generated for every control point along the blade span. Once the horseshoe vortices are defined, the induced velocity at each control point, due to all vortices, is calculated using the Biot-Savart law. The process begins with an initial guess for the circulation values, which are then updated iteratively until the solution converges. This process is illustrated in a flowchart presented in [Figure 2](#).

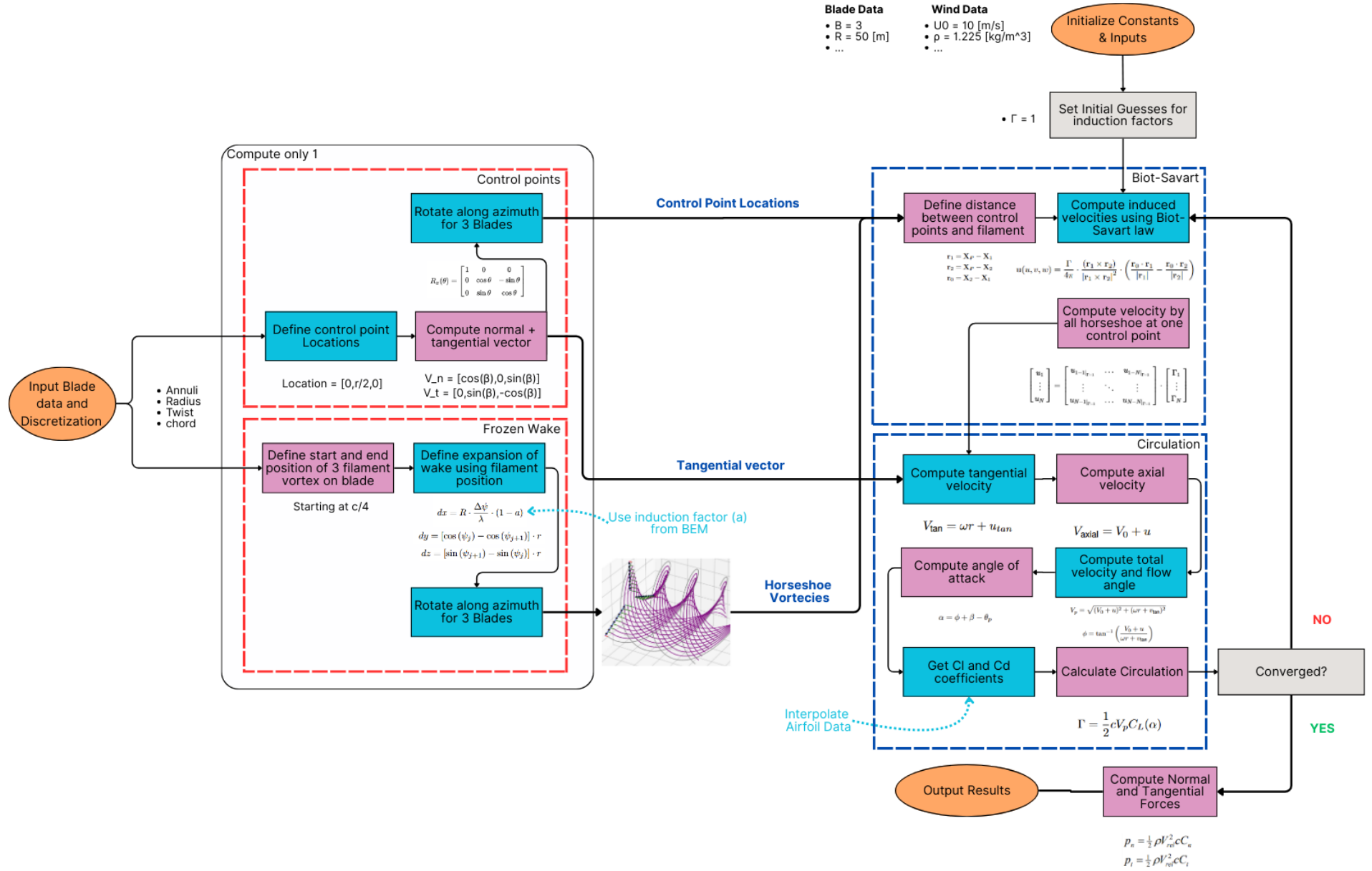


Figure 2: Flow Chart of Lifting Line Model

3 Main Assumptions

To model the performance of a wind turbine, certain assumptions are made in the lifting line model. These assumptions can be separated into two categories. Firstly, the assumptions made in the lifting line theory and secondly, the simplifying assumptions related to the flow case around the wind turbine.

3.1 Assumptions: Lifting Line Model

- **Helmholtz Vortex Theorem:** Vortex filaments conserve their strength (circulation) along their length and cannot terminate within the fluid, they must either form closed loops or extend to infinity. In the lifting line model, this justifies representing each blade with a bound vortex and modeling the wake using trailing filaments that form a continuous horseshoe vortex.
- **Frozen Wake Assumption:** The wake geometry is assumed to be fixed in space. It does not move or deform in any way with time. This makes the aerodynamics quasi-steady instead of unsteady. The vortex sheets formed downstream of the vortex form helical sheets in the wake, which are generated by the blades' rotation and shed vorticity.
- **Wake Convection Velocity:** The wake convects downstream at a speed based on the freestream velocity and an assumed axial induction factor.
- **Helical Trailing Vortices:** The wake of the turbine is modelled as a series of helical filaments shed from the trailing edge of the blades.
- **Blade Modeled as Lifting Line:** Each blade is represented as a line at the quarter chord point, instead of a surface. This assumption makes the bound circulation along the span much simpler than in practice.
- **High Aspect Ratio Blades:** The theory of induced velocity is more accurate for long, slender blades, where the velocity is predominantly in the spanwise direction. This assumption is valid for wind turbine blades based on their shape.
- **Airfoil Data Linear Interpolation:** The data for the lift and drag coefficients at each angle of attack is linearly interpolated from the provided polars of the DU airfoil. This relies on the behaviour of the airfoil for the gap between the two closest angles of attack to be linear. This is simply an approximation and therefore leads to inaccuracies.

3.2 Assumptions: Potential Flow

The lifting line model is based on potential flow, which assumes the following:

- **Uniform Inflow:** The flow velocity is constant $U_0 = 10$ [m/s] across the entire swept area A . This is often not true in real life due to variations in wind speed and direction caused by turbulence, terrain effects, and atmospheric boundary layers. However, the assumption is necessary for potential flow [1].
- **Inviscid:** Viscosity is disregarded, such that boundary layer separation and stall are not modelled and are inferred indirectly through the empirical airfoil data. In real life the presence of viscous leads to boundary layer formation, separation, and potential stall, especially at higher angles of attack [1].
- **Incompressible:** Wind turbines operate below 0.3 Mach and so compressibility effects can be neglected, which is often the case in real life for typical design tip-speed ratios.
- **Steady:** Potential flow assumes a steady, time-independent flow [1]. This is often not true in real life, but the assumption can be made to compute approximate loads experienced.
- **Irrotational:** The flow outside the wake is assumed to be irrotational such that it can be represented by a potential function.

The lifting line theory does not account for any structural deformation due to aeroelastic interactions caused by the exerted forces. When the air exerts a force on rotor blades, they bend due to elastic effects and result in different aerodynamic properties by changing the angle of attack. These phenomena are disregarded in this method. To model more realistic turbulent wind conditions, as well as the flexibility effects of the rotor blades, CFD, including turbulence modelling and fluid-structure interaction, should be implemented.

4 Single Rotor Analysis

4.1 Spanwise Distribution of Angle of Attack and Flow Angle

Figure 3 and Figure 4 illustrate the variation in angle of attack (α) and inflow angle (ϕ) along the rotor blade span for three different tip speed ratios (TSRs): $\lambda = 6, 8$, and 10 .

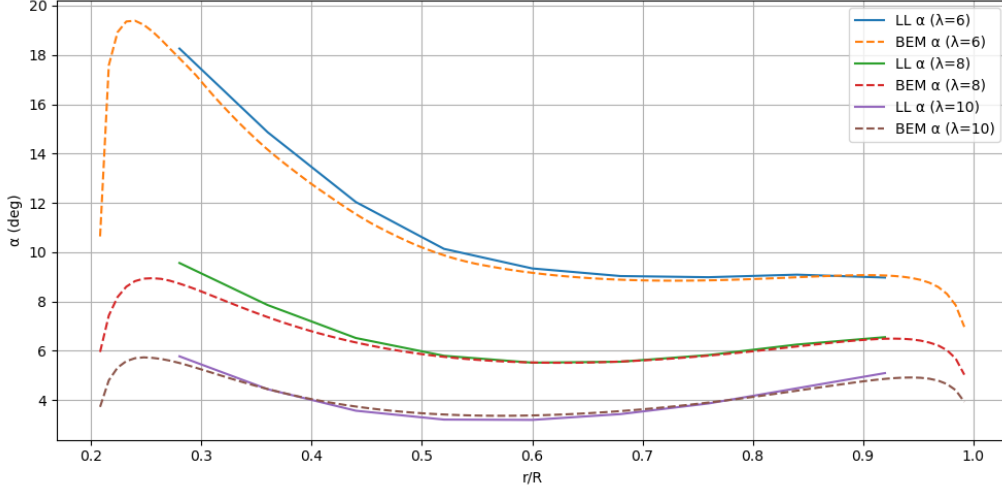


Figure 3: Spanwise distribution of angle of attack α for tip speed ratio $\lambda = 6, 8, 10$.

As seen in Figure 3, the angle of attack generally decreases from root to tip. This trend is primarily due to the increasing local blade speed with radius, which alters the relative flow vector approaching the blade. At higher TSRs, this effect is more pronounced as the flow becomes increasingly aligned with the blade chord, resulting in smaller angles of attack. Lower angles of attack at higher TSRs are typically desirable for avoiding flow separation and reducing drag, which enhances efficiency.

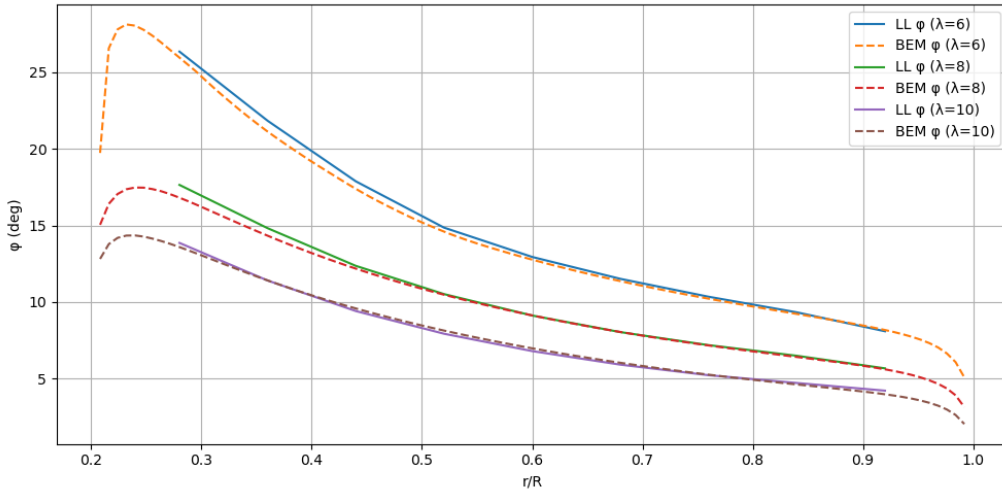


Figure 4: Spanwise distribution of inflow angle ϕ for tip speed ratio $\lambda = 6, 8, 10$.

Figure 4 shows the corresponding spanwise distribution of inflow angle ϕ . This angle, defined as the sum of the local angle of attack α and blade pitch θ and twist β angles, is given by:

$$\phi = \alpha + \theta + \beta \quad (1)$$

Since θ is fixed for the entire blade and β decreases with span, the trend in ϕ largely mirrors that of α . Inflow angle ϕ reflects the orientation of the relative wind vector with respect to the rotor plane. As the tip speed ratio increases, the tangential component of the relative velocity dominates, causing ϕ to decrease. This reduction in ϕ results in a more aligned and smoother flow over the blade sections, reducing aerodynamic loading.

Both the lifting line theory and BEM methods are used to compute the distributions shown in the figures. Across most of the blade span, results from both methods are in close agreement, validating the consistency and reliability of the aerodynamic models. It is also important to note that data for regions very close to the root and tip are absent for the lifting line model. These segments have been intentionally excluded due to the absence of control points in the simulation, which can lead to numerical instabilities and unphysical results in those areas. This will be the case for all of the graphs from here on.

4.2 Spanwise Distribution of Axial and Azimuthal Loads

The aerodynamic forces acting on each blade element can be decomposed into two orthogonal components: the axial force (aligned with the rotor axis) and the azimuthal force (tangential to the rotor sweep, contributing to torque). These are computed using the lift and drag forces, and the local flow angle ϕ , as shown below:

$$p_n = L \cos(\phi) + D \sin(\phi) \quad (2)$$

$$p_t = L \sin(\phi) - D \cos(\phi) \quad (3)$$

The thrust loading p_n , shown in Figure 5, increases with the radius of the blade and peaks near the tip at roughly $r/R \approx 0.9$. Increased TSR shifts increases the normal force significantly, highlighting the influence of rotational speed on loading distribution.

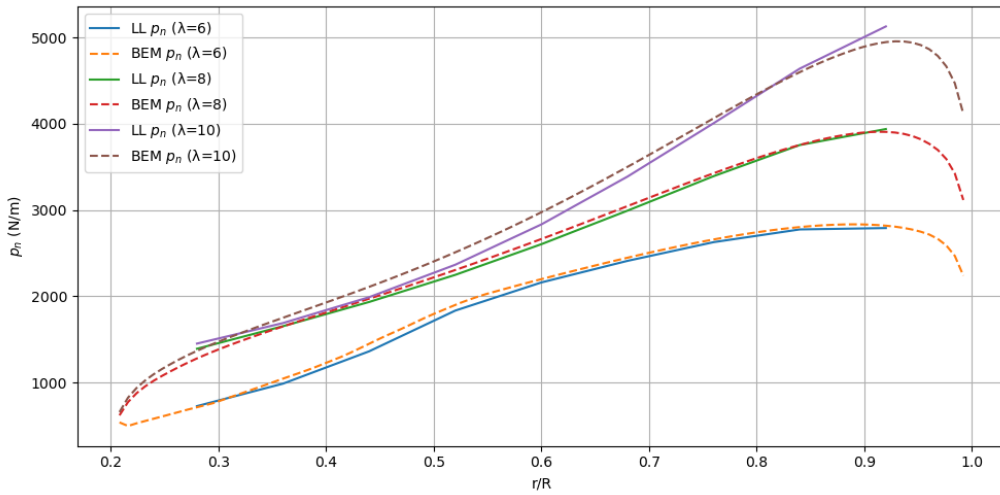


Figure 5: Spanwise distribution of thrust loading p_n for tip speed ratio $\lambda = 6, 8, 10$.

The azimuthal load p_t , shown in Figure 6, directly contributes to torque and power production. It remains relatively constant across the mid-span of the blade, particularly at higher tip speed ratios, where the flow is more aligned with the blade and aerodynamic performance is more stable.

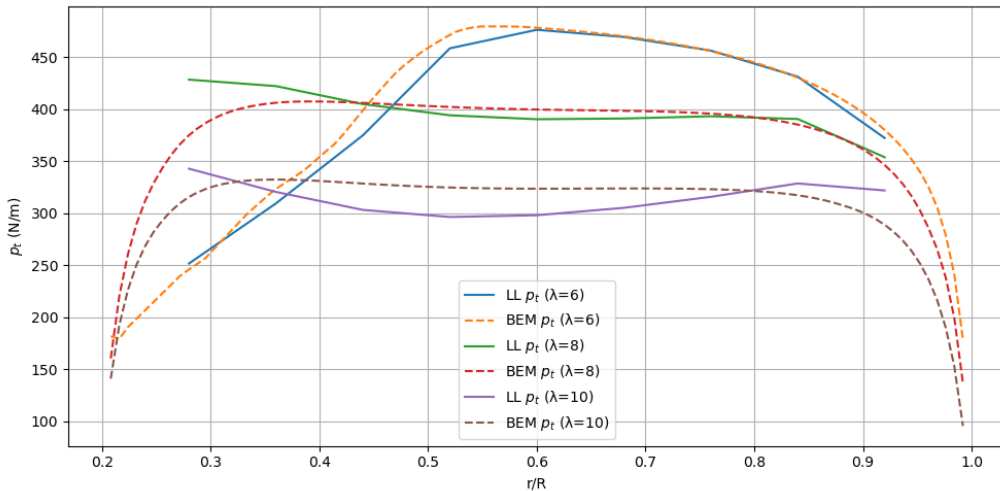


Figure 6: Spanwise distribution of azimuthal loading p_t for tip speed ratio $\lambda = 6, 8, 10$.

When comparing the blade loading between the lifting line model and the BEM method, the agreement is generally strong across most of the span. However, at $\lambda = 10$, some deviations appear near the mid-span region. These discrepancies could be due to differences in how each model handles induced velocities and three-dimensional flow effects at higher rotational speeds. In particular, the lifting line method may slightly underestimate the induced inflow depending on the spatial resolution or wake modeling assumptions, leading to local variations in the computed thrust and azimuthal loads.

4.3 Spanwise Distribution of Circulation

Circulation Γ at a given blade element represents the rotational flow, or vorticity, around the airfoil section. To compute the spanwise distribution of circulation, the lift per unit span $L' = \frac{1}{2}\rho V_{rel}^2 C_l c$ is used to derive the following expression:

$$\Gamma = \frac{1}{2} V_{rel} C_l c \quad (4)$$

Implementing Equation 4 in the lifting line code yields the circulation distribution along the blade, as shown in Figure 7. Note that the circulation has been normalised by dividing by $\frac{\pi V_0^2}{B(\lambda V_0/R)} = \frac{\pi V_0^2}{B\omega}$.

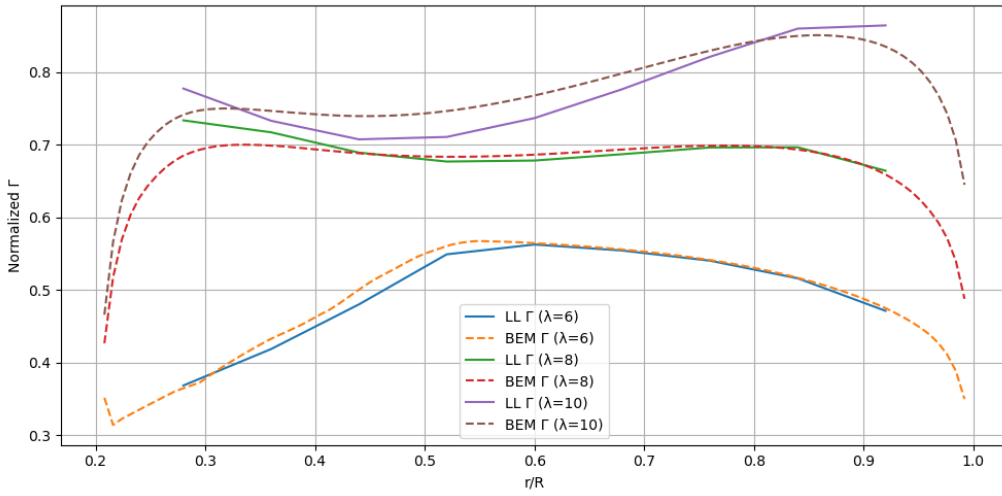


Figure 7: Spanwise distribution of circulation Γ for tip speed ratio $\lambda = 6, 8, 10$.

Circulation remains relatively constant in the mid-span of the blade for the higher TSRs and drops toward both the root and tip. This is consistent with the Prandtl lifting line theory. As the tip speed ratio increases, the circulation distribution becomes more uniform along the span, indicating more efficient lift generation. The general circulation trend remains consistent across different λ values, except near the root for $\lambda = 6$, where stall is likely responsible for a drop in circulation and thus lift coefficient. According to Helmholtz's vortex theorems, circulation is directly linked to the vorticity shed into the wake, which must be conserved. This explains the strength and structure of trailing vortices generated by the blade.

The lifting line circulations seem to match up very well with those from BEM, even capturing the drop in circulation close to the root for $\lambda = 6$. The biggest difference is seen in the mid-span for $\lambda = 10$, which could explain the slight deviations in thrust and azimuthal loading for these conditions as was shown in Figure 5 and Figure 6.

4.4 Total Thrust and Torque

Lastly, Figure 8 presents the global performance metrics for the wind turbine; namely the power coefficient C_P and thrust coefficient C_T . The power coefficient is seen to increase with tip speed ratio up to a point, indicating improved efficiency, but plateaus at $\lambda = 10$ and may drop beyond the optimal range due to increased drag as TSR continues to increase. The thrust coefficient also increases with TSR, implying greater axial force on the rotor. This can have structural implications if the TSR becomes too high, requiring stronger blades with higher stiffness to withstand the loads.

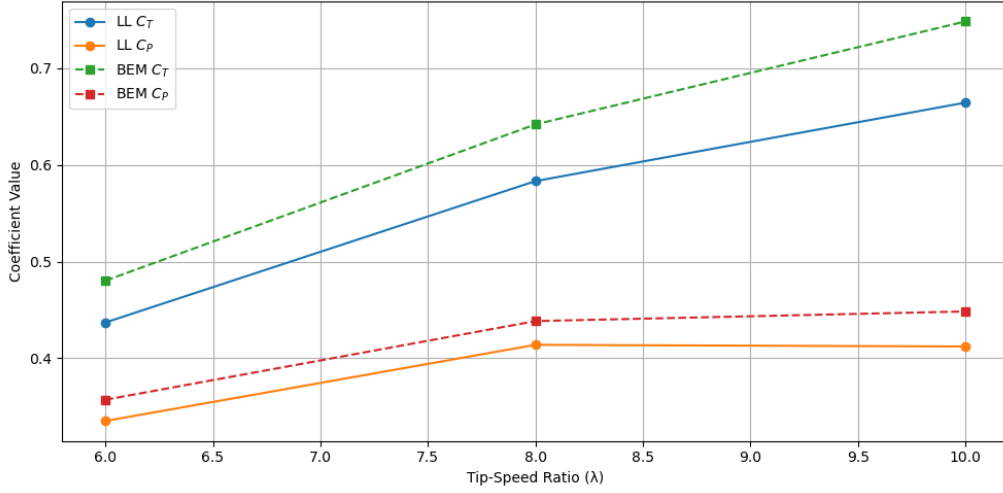


Figure 8: Power and thrust coefficients C_P , C_T for tip speed ratio $\lambda = 6, 8, 10$.

Once again, there seems to be a slight deviation between the lifting line and BEM results, with the former underestimating both C_P and C_T slightly. The main reason for this difference is the fact that the lifting line control points do not reach as close to the root and tip of the blade as BEM does. This was done on purpose since placing control points too close to the edges lead to numerical instability and non-sensical results. Consequently, the power and thrust generated close to these edges is not captured by the lifting line approach, creating a deficit in comparison to BEM.

5 Impermeability Boundary Condition

The impermeability boundary condition can be used as an alternative method to solve for circulation Γ , without needing iterations to achieve convergence. This boundary condition states that the flow normal to the blade's surface is zero. Furthermore, Prandtl's lifting line theory states that the velocity at a given point consists of the components from vortices bound by the blade w_b , the wake w_i and the freestream velocity U_∞ . In the normal direction this can be expressed as:

$$w_i + w_b + U_\infty \cdot \alpha = 0 \quad (5)$$

To satisfy the impermeability boundary condition, a second set of points, called the collocation points, is set at 3/4 of the chord length according to the Pistoletti theorem. These points serve as the locations at which the velocity in the normal direction is zero. Consider Figure 9 which indicates the set-up for this new approach and Figure 10, which shows the control and collocation points for the turbine being analysed. The distance between the control and collocation points vary along the span due to the changing chord of the blade.

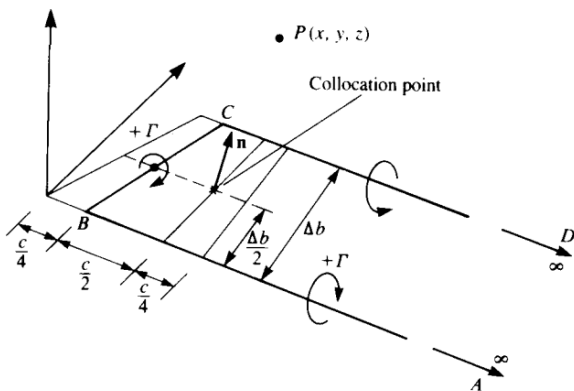


Figure 9: A spanwise horseshoe vortex indicating a collocation point at 3/4 chord. Taken from [3].

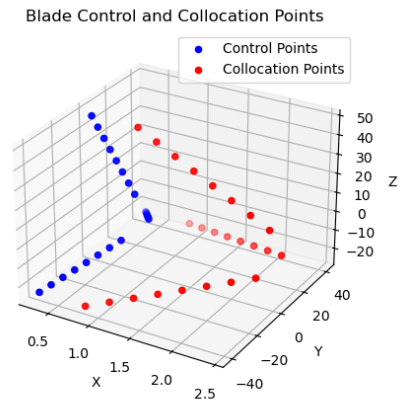


Figure 10: Layout of control points and collocation points.

Using the impermeability boundary condition, the system of linear equations presented in Equation 6 is formulated; where u, v, w are the velocities induced by the vortex segments.

$$(u, v, w)_{ij} \cdot \mathbf{n}_i \Gamma_i = -(U_\infty, V_\infty, W_\infty) \cdot \mathbf{n}_i \quad (6)$$

This system can then be solved to determine the circulation along the blade span without iterating. The circulation can be linked to the lift through the Kutta-Joukowski theorem, as seen in Equation 4, to be used in further calculations regarding the performance of the wind turbine as was done in section 4. The impermeability boundary condition approach is illustrated in the flow chart in Figure 12, which highlights the differences in how circulation Γ is calculated compared to the iterative method shown in Figure 2.

The spanwise circulation distributions computed using the two lifting line methods are compared with the BEM results in Figure 11. At a tip speed ratio of $\lambda = 8$, the boundary condition method shows strong agreement with both the iterative lifting line solution and the BEM reference, indicating that the approach can produce accurate results under favorable flow conditions. However, at the lower and higher tip speed ratios ($\lambda = 6$ and $\lambda = 10$), the boundary condition method displays notable deviations. Across much of the blade span, it tends to overpredict circulation compared to the other methods. In the outer portion of the blade ($r/R > 0.75$), this trend reverses, and the method begins to underpredict circulation. This shift suggests a sensitivity in the model's ability to accurately resolve the induced velocity field near the blade tip, where three-dimensional flow effects and tip losses become more pronounced.

The primary suspected source of error lies in the definition of the normal vectors at the collocation points. Since the impermeability condition is enforced by setting the velocity component normal to the blade surface to zero, even small inaccuracies in the orientation of the normal vectors can lead to incorrect estimates of the induced velocities. These errors, in turn, affect the computed circulation values and can propagate across the span, particularly in regions where the flow is highly sensitive to wake interactions. Improving the geometric accuracy of the normal vector definition or refining the collocation point placement may help reduce these discrepancies.

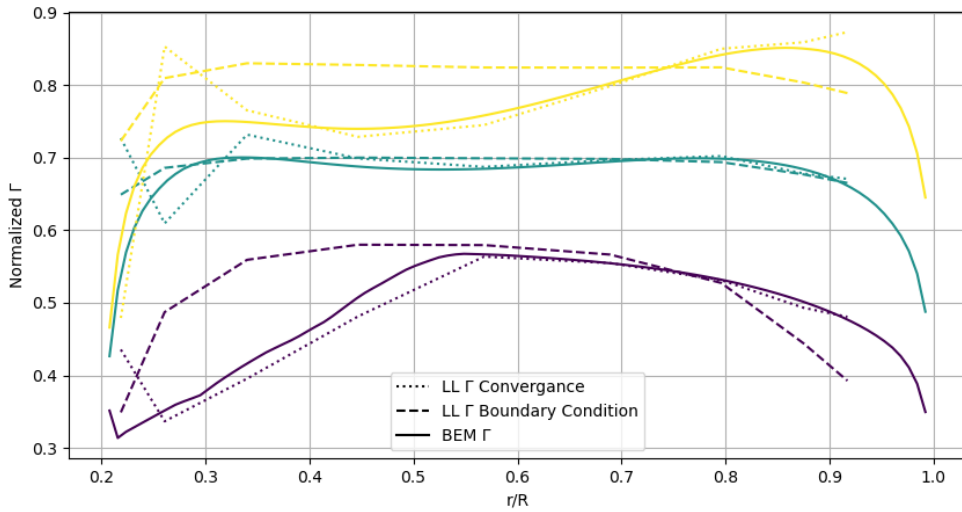


Figure 11: Circulation computed with convergence method and impermeability boundary condition, compared to results from BEM presented in [2], for $\lambda = 6$ (purple), 8 (blue/green) & 10 (yellow).

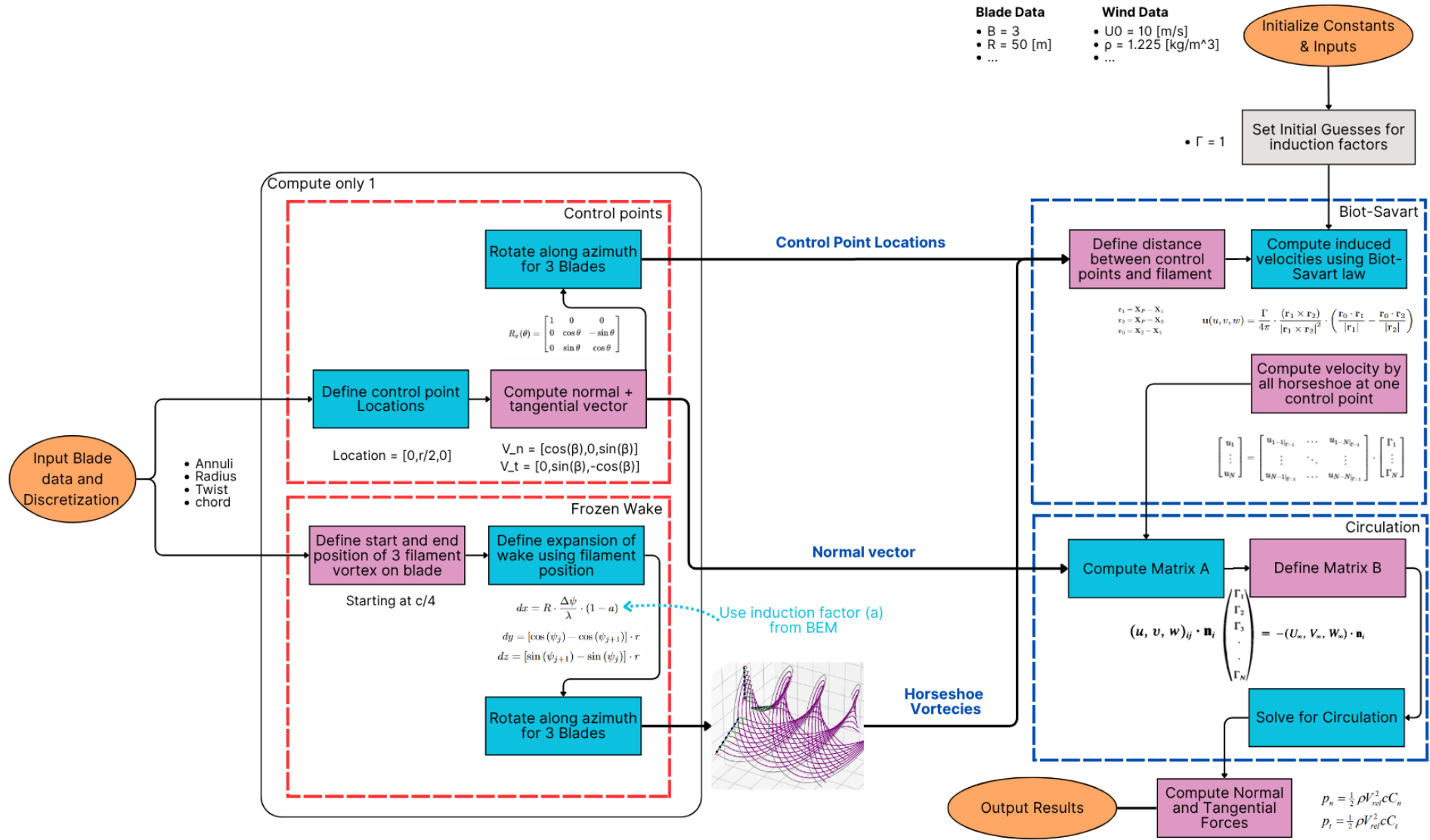


Figure 12: Flow Chart of Lifting Line model, using impermeability boundary condition to solve for circulation.

6 Multi Rotor Analysis

6.1 Model configuration and description

A second rotor was modelled next to the original at distances $L = [D, 2D, 5D, \infty]$ from each other along the y-axis. Furthermore, to evaluate the effect of a rotation phase difference between rotors, a phase angle was applied to the second rotor in increments of 30° . This configuration can be seen in Figure 13, where the second rotor is placed at $2D$ from the first and has been rotated by 60° . For representational and convenience purposes, it was assumed that $L = \infty \approx 30D$.

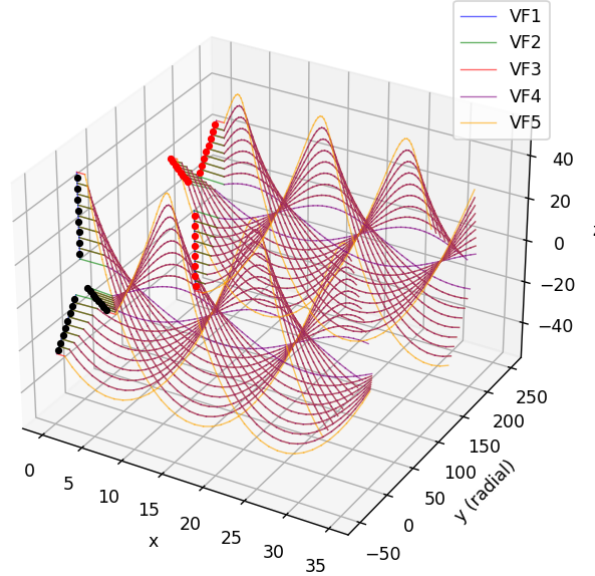


Figure 13: *Frozen wake helix made of filaments for the multi rotor analysis*

The multi-rotor analysis was performed using ten nodes to maintain coherence with the single-rotor analysis; however, due to intensive computational costs, the number of rotations for the frozen wake model was limited to one. The effects of this are discussed in detail below in subsection 7.4, but since the intended objective is to compare the results against each other, these effects can be disregarded for now as they remain part of the control parameters. Another parameter that was held constant for the multirotor analysis is the tip speed ratio, which was kept at six. Additionally, the circulation was computed using the convergence method.

6.2 Evaluation of rotor spacing on performance

The performance of the rotors was assessed under these circumstances for each rotor spacing L and plotted in the figures below.

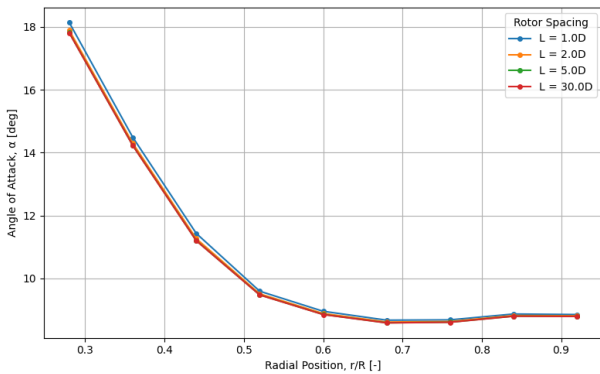


Figure 14: *Radial Distribution of angle of attack α at Phase angle = 0° and $\lambda = 6$*

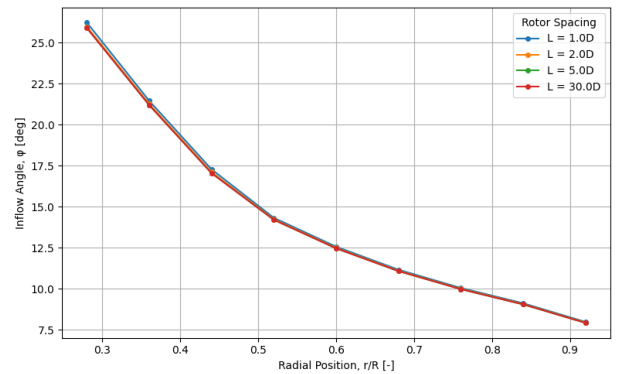


Figure 15: *Radial Distribution of inflow angle ϕ at Phase angle = 0° and $\lambda = 6$*

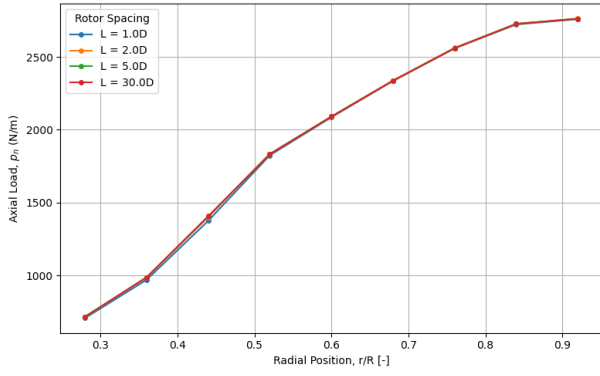


Figure 16: *Radial Distribution of axial load p_n at Phase angle = 0° and $\lambda = 6$*

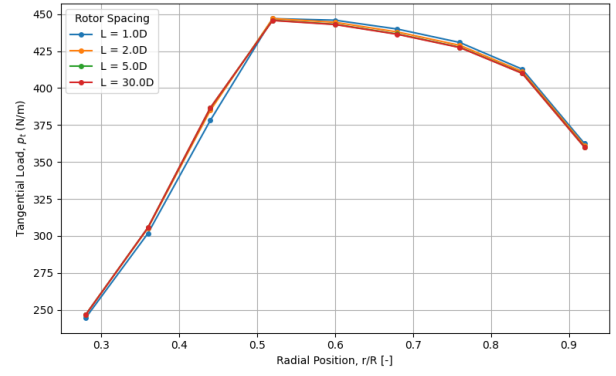


Figure 17: *Radial Distribution of tangential/azimuthal load p_t at Phase angle = 0° and $\lambda = 6$*

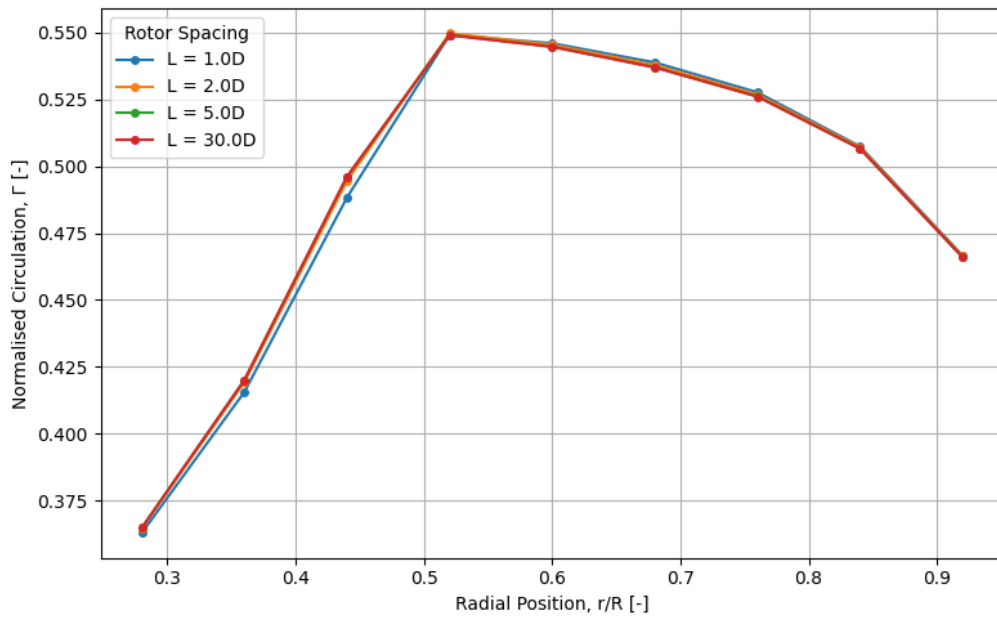


Figure 18: *Radial Distribution of normalised circulation Γ at Phase angle = 0° and $\lambda = 6$*

These figures demonstrate how the presence of a second rotor nearby has little impact on the performance of the primary rotor. As expected, the greatest disparity arises from comparing the closest with the furthest rotor, with the latter being able to be considered independent, as detailed below in [subsection 6.4](#). However, even within the great degree of similarity in results, the angle of attack and inflow angle both appear to be affected more by changes in rotor spacing at the root of the blade than at the tip. This might be a product of the same issues shown and discussed in [subsection 4.1](#). The trends and values shown for the axial and tangential forces match closely with the ones obtained in the single rotor case at the same tip speed ratio, indicating an overall maintenance in performance. However, this can be best visualised by computing the coefficients of power and thrust, which are displayed below in [Figure 19](#).

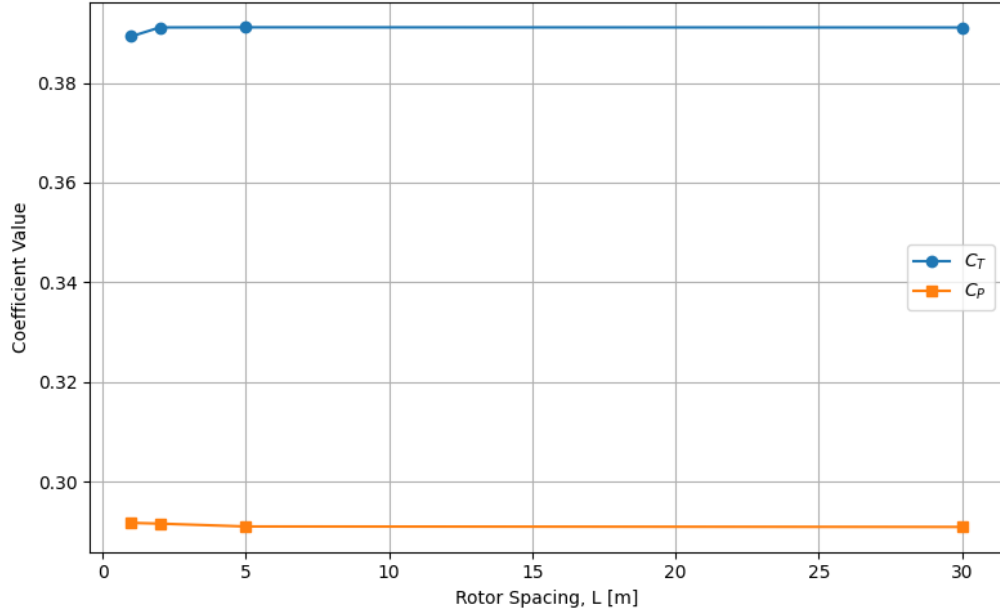


Figure 19: C_T and C_P for increasing rotor spacing (in diameters) at Phase angle = 0° and $\lambda = 6$

Both C_T and C_P exhibit relatively stable behaviour for spacings beyond $L \geq 2D$, indicating that aerodynamic interference between rotors diminishes with increased separation. The decrease in C_P suggests that close proximity increases induced tangential velocity, increasing torque and power extraction, but this effect diminishes with separation, while the increase in C_T suggests that mutual interference at close spacing reduces the effective angle of attack and therefore the lift generated by the blade.

6.3 Evaluation of rotation phase difference on performance

To appropriately evaluate the effect of distance, the effect of the rotation phase difference between rotors must also be considered, and for that purpose, the following figures for the previous parameters were plotted under a constant rotor spacing of D and varying rotation phase angles.

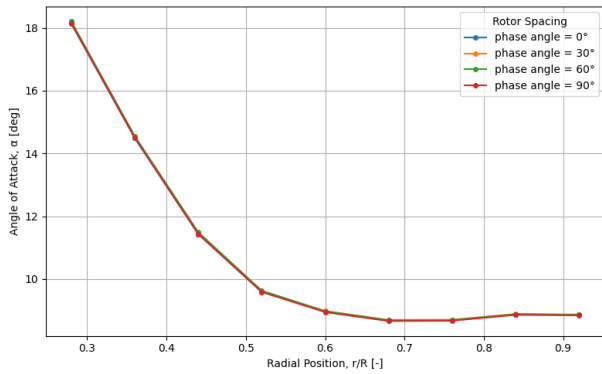


Figure 20: Radial Distribution of angle of attack α at $L = D$ and $\lambda = 6$

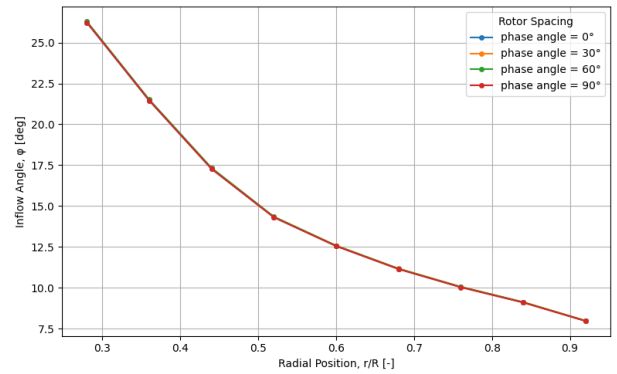


Figure 21: Radial Distribution of inflow angle ϕ at $L = D$ and $\lambda = 6$

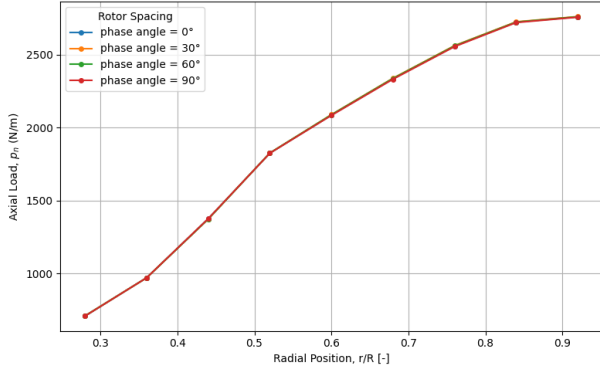


Figure 22: Radial Distribution of axial load p_n at $L = D$ and $\lambda = 6$

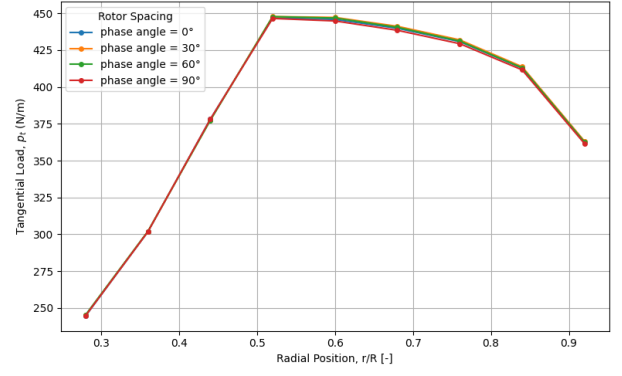


Figure 23: Radial Distribution of tangential/azimuthal load p_t at $L = D$ and $\lambda = 6$

In these figures, the resemblance between values is even greater, with only Figure 23 showcasing any visible discrepancies at the final half of the blade.

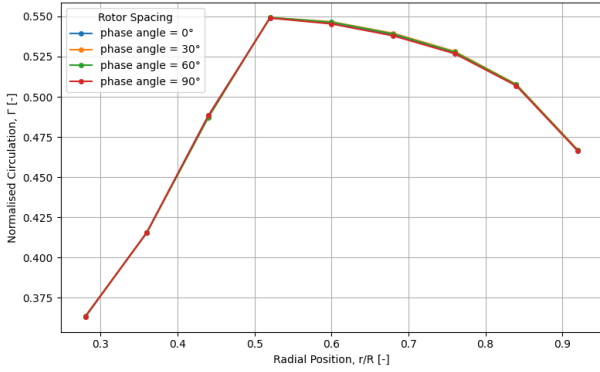


Figure 24: Radial Distribution of normalised circulation Γ at $L = D$ and $\lambda = 6$

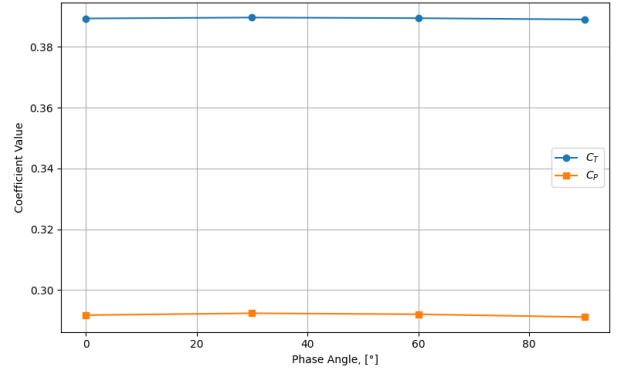


Figure 25: C_T and C_P for increasing phase angle difference at rotor spacing D and $\lambda = 6$

The results for the coefficients indicate that both C_T and C_P remain remarkably stable across the phase range from 0° to 90° . This suggests that, at close spacing, the relative azimuthal alignment of the two rotors has minimal influence on the time-averaged aerodynamic performance. This insensitivity to phase shift implies that although local blade-blade interactions may vary instantaneously with phase, the net inflow conditions over a full revolution are nearly phase-invariant at this spacing. This is consistent with the symmetry of the side-by-side arrangement and highlights that performance predictions in such configurations can be robust even without accounting for phase synchronisation.

6.4 Streamtube independence

This section presents the average normalised circulation difference $\langle |\Gamma_B - \Gamma_A| / \Gamma_A \rangle$ as a function of azimuthal phase angle at a fixed rotor spacing $L = D$.

At zero phase difference, the two rotors operate in phase, and the mutual aerodynamic interference is strongest, as indicated by the highest relative circulation difference. As the phase angle increases, the interaction weakens slightly, and a sharper reduction is observed at 90° . This trend suggests that streamtube independence improves with phase offset, likely due to phase-induced cancellation or redistribution of induced velocities between blades. The results indicate that while phase difference has limited influence on performance coefficients as seen in Figure 25 it does affect the spatial distribution of circulation and wake interaction, particularly near maximum phase offset.

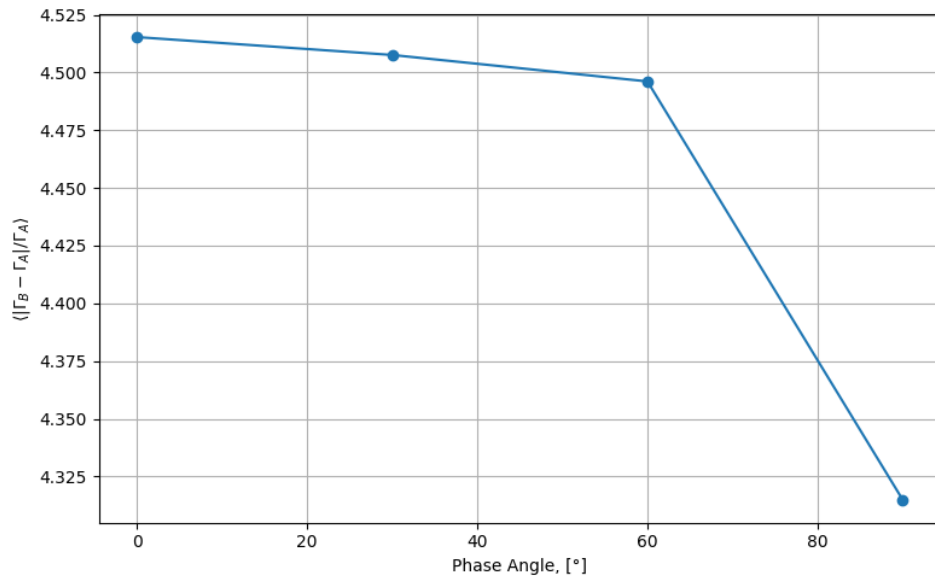


Figure 26: Average normalised circulation difference as a function of phase angle

It is worth noting that for the wake geometry in the two-rotor configuration, the analysis and evaluations are performed between blade 1 of both rotors. Were the full rotor to be considered, the interaction between blade 1 at 90° and the unphased blade 2 would be equivalent to the interaction between a 30° phased blade 1 with the other unphased blade 1. This means that for the whole rotor, the configuration with the least interference would be under a 60° phase difference.

7 Sensitivity Studies

To assess the sensitivity of the lifting line model to various input parameters, a sensitivity analysis was conducted. The analysis focuses on evaluating changes in both normal and azimuthal blade loading at a fixed tip speed ratio of $\lambda = 8$. For reference and validation, the results are compared against those obtained from the BEM method.

7.1 Convection Speed for Frozen Wake

Firstly, the effect of the convection speed of the wake will be evaluated. This can be modelled through the use of the axial induction factor a , where:

$$U_{\text{wake}} = U_{\infty}(1 - a) \quad (7)$$

Higher values of the axial induction factor a correspond to lower wake velocities, indicating greater energy extraction and therefore a higher thrust force generated by the rotor. Figure 27 shows that as a increases from 0 to 0.3, the blade loading decreases slightly. Notably, the case with $a = 0.3$ produces results that most closely match the BEM solution, with an almost identical distribution of loads.

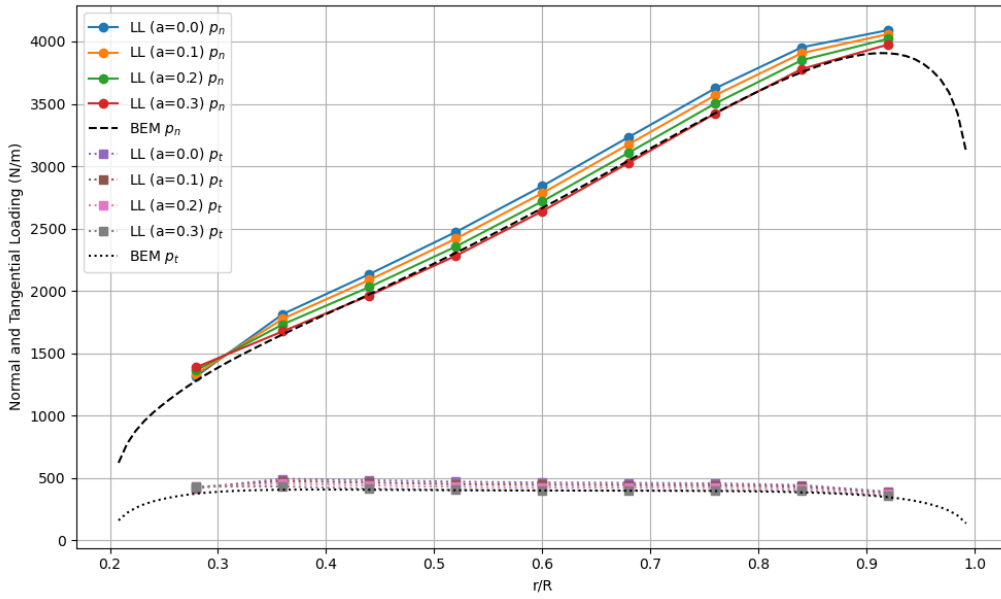


Figure 27: Effect of axial induction factor a on blade loading for $\lambda = 8$.

This behavior arises because the axial induction factor directly affects the spacing and strength of the trailing wake vortices in the frozen wake model. A higher a value increases the influence of the induced wake velocity field on the control points, resulting in slightly reduced aerodynamic loading. The closer agreement with BEM at $a = 0.3$ suggests that this value offers a more physically representative approximation of wake convection in the lifting line framework for this particular operating condition.

7.2 Discretization of the Blade

The influence of blade discretization on the modelled blade loading is illustrated in Figure 28. Two discretization types are evaluated; uniform spacing and cosine spacing, where control points are clustered toward the tip and root. The figure shows that across the central region of the blade, there is little difference between both types of spacing. However, near the blade root and tip, cosine spacing extends the modeled region further than uniform spacing and achieves better alignment with the BEM results.

That said, as noted previously, placing control points too close to the blade extremities can introduce numerical instabilities in the lifting line model, as can be seen from the cosine spacing. These instabilities may result in unrealistic or non-physical load predictions, which must be treated with caution during interpretation.

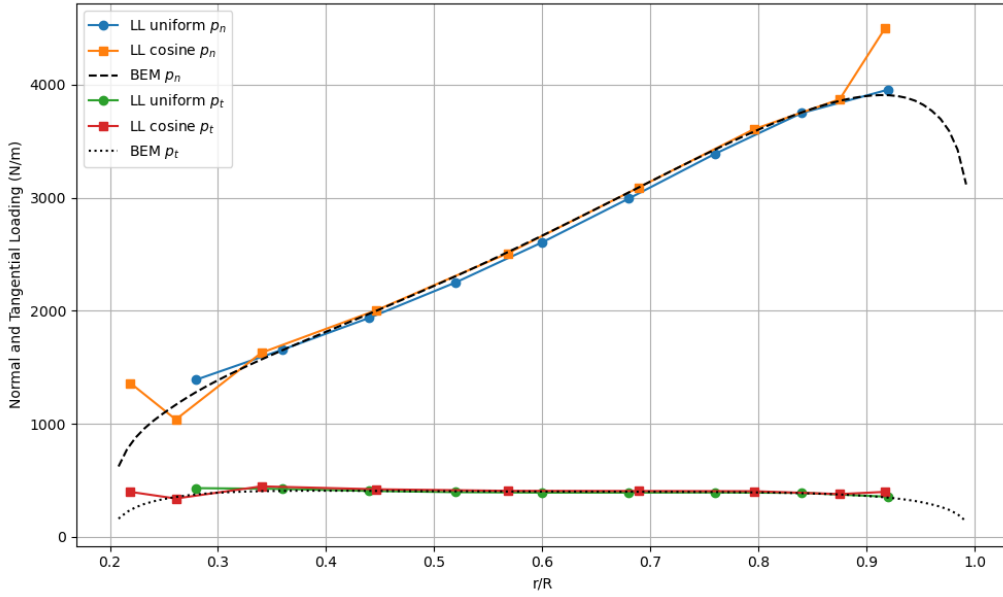


Figure 28: Effect of blade discretization type (uniform, cosine) on blade loading for $\lambda = 8$.

7.3 Azimuthal Discretization

Another important parameter influencing the accuracy of the lifting line model is the number of azimuthal discretization points used as the turbine rotates, denoted as N_{azim} . Figure 29 illustrates how varying this number affects the spanwise distribution of the blade loading at a fixed tip speed ratio of $\lambda = 8$.

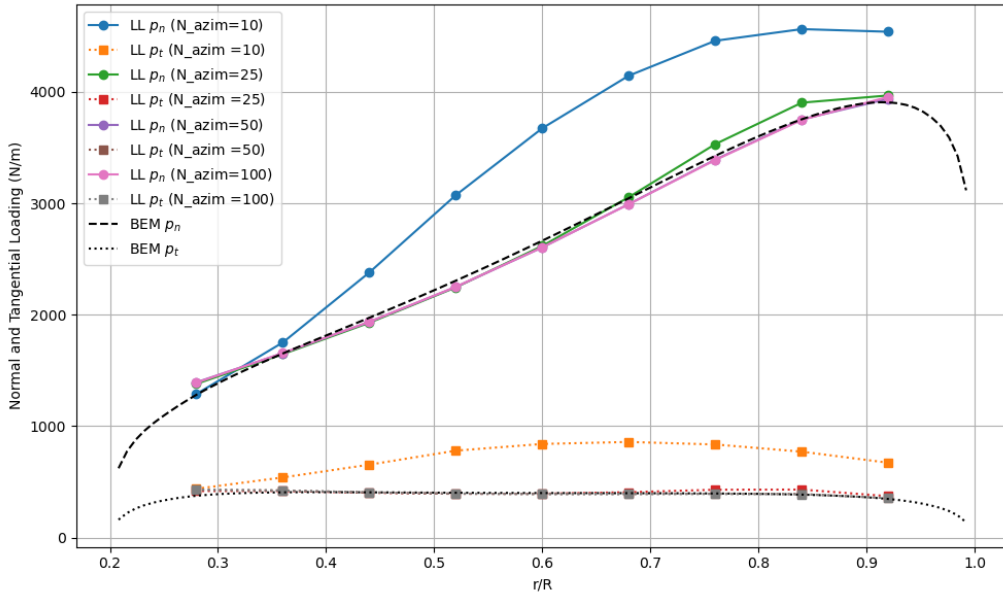


Figure 29: Effect of azimuthal discretization on blade loading for $\lambda = 8$.

The plot shows that at low values of N_{azim} (e.g. 10), the computed loads significantly deviate from the BEM reference solution for both the tangential and normal directions. As N_{azim} increases to 25, 50, and 100, the agreement with BEM improves considerably.

Increasing azimuthal resolution enhances the fidelity of the wake interaction modeling, allowing for more accurate computation of the induced velocity field at each control point. However, this improvement comes at the cost of increased computational demand. Based on the results, values of $N_{\text{azim}} = 50$ appear sufficient for convergence and good agreement with BEM predictions.

7.4 Length of the Wake

Lastly, the effect of wake length is examined by varying the number of rotor revolutions used to model the wake. A greater number of rotations corresponds to a longer wake, representing a larger time span over which the shed vorticity influences the flow. As shown in Figure 30, the circulation distribution resulting from a wake length of just one rotation deviates significantly from the BEM solution. However, when the wake length is extended to 5 or more rotations, the results closely match the BEM predictions across most of the blade span.

Interestingly, at around $r/R \approx 0.85$, the case with 20 rotations begins to diverge slightly more from the BEM results than the 5 or 10 rotation cases. This small deviation may be attributed to numerical accumulation of errors or increased sensitivity to discretization at the blade tip, where the influence of the extended wake becomes more complex and less stable.

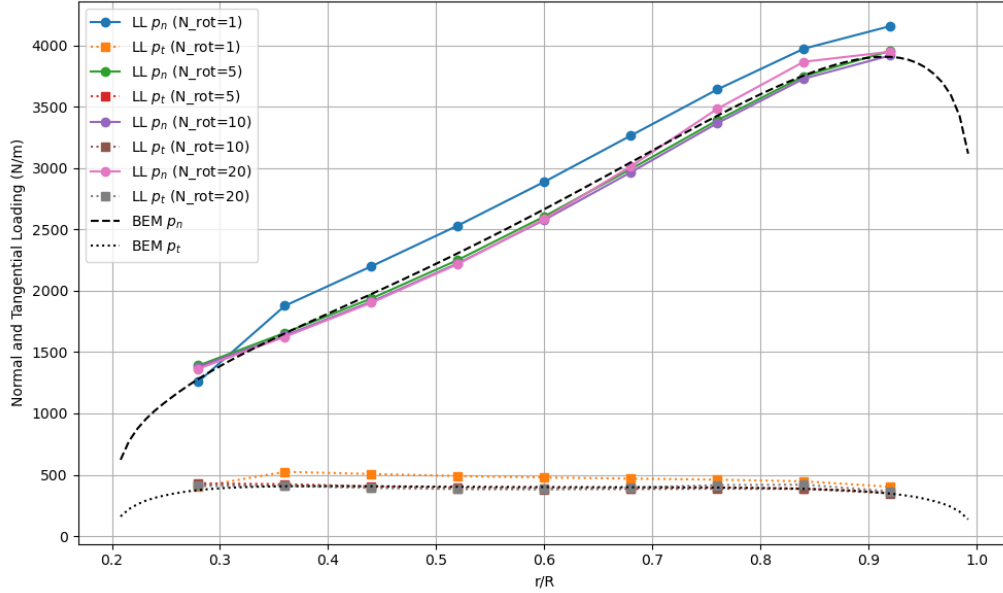


Figure 30: Effect of number of rotations (length of wake) on blade loading for $\lambda = 8$.

In addition to accuracy, the number of iterations required for circulation to converge can be evaluated for different wake lengths, as shown in Figure 31. The general trend indicates that longer wakes require more iterations to reach convergence. This is likely because a greater number of wake filaments contributes to the induced velocity at each control point, increasing the complexity of the interaction and slowing convergence. However, the case with 20 rotations unexpectedly converges faster than the cases with 5 or 10 rotations. This unusual behavior could potentially be due to numerical smoothing or damping effects that occur when many distant wake filaments have relatively small and evenly distributed contributions. As a result, their combined influence may lead to a more stable circulation update in each iteration.

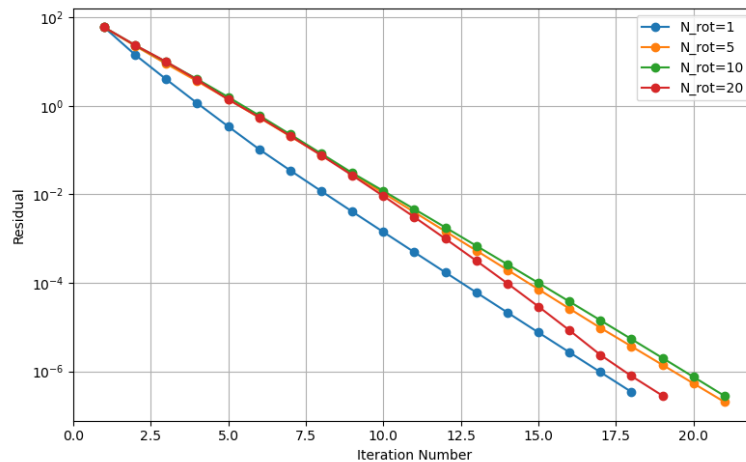


Figure 31: Convergence of circulation for varying wake length.

8 Discussion and Conclusion

The lifting line theory model demonstrated in this paper indicates comparable results with BEM for spanwise distributions. Loads in the axial and azimuthal loads, as well as the flow angle and angle of attack, show strong correlations across the three different tip speed ratios. The clear drawback of the comparison between these two methods is near the extremes of the rotor blades. Due to the location of the control points, the lifting line model struggles to accurately capture the aerodynamics at the root and the tip of the blade. This results in underestimated thrust and power coefficients, as the loads are integrated over the length of the rotor.

Sensitivity analyses presented indicate a dependence on the induction factor a that describes how the frozen wake expands downstream. This dependence is a significant drawback of this model as it requires an adequate value of a to resemble BEM results. Further analysis also showed that more rotations in the wake than five are not necessary, as the result tends to the same value, saving some computational cost. The accuracy of the results does benefit from a smaller azimuthal discretisation, especially near the edges of the rotor.

To conclude, the lifting line model is valid and yields results similar to those of BEM. Some caution should be used when implementing the impermeability boundary condition, which has been shown to provide varying and inconsistent results. Although the general shape and magnitude of the output circulation are within reasonable limits, the general shape is not represented adequately. Based on the findings of this report, solving for circulation by convergence is recommended instead.

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